When Does a Variational Quantum Classifier Head Help?

An Empirical Comparison Against Capacity-Matched Classical Heads on MNIST and CIFAR-10

Alexander Greene
alexanderngreene@gmail.com

Abstract

Hybrid quantum–classical models, in which a classical neural network feeds a variational quantum circuit (VQC), are a leading candidate architecture for near-term quantum machine learning. Yet controlled comparisons against classical baselines of comparable parameter count remain scarce, and results on datasets beyond MNIST are rarer still. We train a shared convolutional backbone with three interchangeable classifier heads, namely a linear layer, a near-parameter-matched multilayer perceptron (MLP), and an 8-qubit VQC, on stratified low-data subsets of MNIST and CIFAR-10, with three random seeds per configuration. On MNIST, the VQC head outperforms the near-matched MLP at every training-set size, by roughly 7–18 percentage points, and trails a smaller linear reference head by only 6–8 points. On CIFAR-10 the ordering inverts: the VQC falls behind both classical heads at every training-set size; the deficit narrows as data is added but remains large. Notably, the VQC never outperforms the simple linear head in any condition. Data re-uploading degrades accuracy in every configuration tested, with the strongest re-uploading schedule driving the circuit into underfitting (train–validation gap <1 point). Our results suggest that the dominant constraint as image complexity grows is not primarily optimization difficulty but the fixed rotation-angle encoding applied to a narrow bottleneck: the quantum head is competitive precisely when the bottleneck embedding is already nearly linearly separable.

1 Introduction

Variational quantum circuits (VQCs) trained by gradient descent are the workhorse of near-term quantum machine learning [1, 2]. In image classification, the dominant hybrid pattern delegates feature extraction to a classical convolutional network and classification to a small quantum circuit [3, 4]. Reported successes, however, are difficult to interpret: quantum heads are frequently compared against classical baselines with different parameter budgets, different backbones, or no controlled seed replication, and evaluation is concentrated on MNIST, whose features are nearly linearly separable after even shallow convolutional processing.

This paper asks a narrow question with a controlled experiment: *holding the backbone, the bottleneck dimension, and the optimizer fixed, and matching the head parameter count as closely as the architectures allow, when does replacing a classical classifier head with a VQC help, and when does it hurt?* We contribute:

1. A controlled comparison of linear, near-parameter-matched MLP, and VQC heads on a shared CNN backbone, across three low-data regimes on both MNIST and CIFAR-10. In total we report 90 training runs: 2 datasets $\times$ 3 training-set sizes $\times$ 5 head configurations (linear, MLP, and quantum with three re-uploading schedules) $\times$ 3 seeds.

2. Evidence that the VQC head's relative standing depends strongly on dataset complexity: it beats the near-matched MLP on MNIST at every training-set size, loses to it on CIFAR-10 at every training-set size, and never outperforms a smaller linear head on either dataset.

3. An ablation of data re-uploading [5] showing that, in this bottlenecked hybrid setting, every re-uploading schedule tested reduces accuracy, and the most aggressive schedule induces underfitting rather than overfitting.

2 Architecture and Methods

Shared backbone. All heads consume the same embedding: a CNN backbone ends in global average pooling followed by a linear projection to a $d = 8$ dimensional bottleneck. For MNIST the backbone has three convolutional blocks (32–64–128 channels, ∼94K parameters); for CIFAR-10, four blocks (64–128–256–256 channels, ∼0.96M parameters). The backbone is trained jointly with the head in every run; gradients flow through the quantum circuit into the convolutional layers.

Quantum head. The bottleneck vector $\mathbf{x} \in \mathbb{R}^8$ is encoded into an 8-qubit circuit by single-qubit rotations $R_y(\pi \tanh x_i)$ on qubit i. The variational block repeats $L = 4$ layers, each applying learned $R_y(\theta)\, R_z(\phi)$ rotations per qubit followed by a ring of CNOT entanglers. The circuit outputs $\langle Z_i \rangle$ for each qubit, and a final classical linear layer maps the 8 expectation values to 10 class logits. The head totals 154 trainable parameters: $8 \times 2 \times 4 = 64$ rotation angles plus $8 \times 10 + 10 = 90$ readout weights. Rotation angles are initialized from $\mathcal{U}(-0.1, 0.1)$, keeping the initial state close to $|0\rangle^{\otimes 8}$ where gradients are well-conditioned; we found uncontrolled initialization to be a dominant source of seed variance. Circuits are simulated with PennyLane's `lightning.qubit` statevector backend using adjoint differentiation [8].

Data re-uploading. We test three encoding schedules. NONE encodes $\mathbf{x}$ once, before the first variational layer. LIGHT additionally re-encodes a rescaled copy ($0.5\mathbf{x}$) midway through the variational block. STRONG places one full encoding of $\mathbf{x}$ immediately before *each* of the $L = 4$ variational layers (the first of these serves as the initial encoding; there is no separate extra encoding), following Pérez-Salinas et al. [5].

Classical baselines. The *linear* head is a single dense layer (90 parameters). The *MLP* head is $8 \to 8 \to 10$: one hidden layer of width 8 with ReLU activation, no dropout or normalization, PyTorch-default (Kaiming-uniform) initialization, and the same optimizer and learning rate as every other head (not tuned separately). Its 162 total parameters (weights and biases of both layers) are the closest the architecture comes to the quantum head's 154; we therefore describe the MLP as *near*-parameter-matched rather than equal. The linear head is smaller than both and is included as the strongest simple reference baseline (a linear probe of the embedding), not as a capacity-matched comparison.

Training protocol. All runs use AdamW (head learning rate 10^{-3}, backbone 5×10^{-4}, weight decay 10^{-4}), cosine-annealed to zero over the run, batch size 32, and gradient clipping. Training subsets are stratified samples of the full training set, preserving class balance. MNIST runs train for 25 epochs with clip norm 1.0; CIFAR-10 runs for 80 epochs with clip norm 2.0 and standard augmentation (random crop, horizontal flip, rotation, color jitter). Every configuration is repeated with seeds $\{0, 1, 2\}$; we report the mean and *sample* standard deviation over the three seeds. Per-component gradient norms (backbone vs. head) are logged every epoch to monitor trainability.

Evaluation caveat. Accuracy is measured on the official 10,000-image test split after every epoch, and we report the best epoch per run. Because no separate validation split was held out of the training data, the test split participates in checkpoint selection; we therefore call the metric *test-split accuracy* rather than test accuracy, and we quantify the effect directly: mean final-epoch accuracy is within 1.3 points of best-epoch accuracy in every one of the 30 configurations (within 0.5 points in 26 of them), and no ordering among heads changes if final-epoch accuracy is used instead. A train/validation/test protocol is planned for a future revision.

3 Results

MNIST: the quantum head beats its near-matched baseline but not a linear probe. Table 1 shows the VQC head (NONE) exceeding the 162-parameter MLP by 17.8, 18.0, and 6.9 points at 250, 500, and 1,000 training samples respectively, while trailing the 90-parameter linear head by a roughly constant 6–8 points. The quantum head is thus the better of the two similarly sized nonlinear heads, but it never beats the simplest baseline. At 1,000 samples it is also the most reproducible model in the study (sample std 0.3 across seeds). In this regime the 8-dimensional embedding learned by the backbone is nearly linearly separable (the linear head reaches 89.6%), and the VQC's fixed angular encoding $R_y(\pi \tanh x_i)$ preserves enough of that structure to classify well.

Table 1: Best test-split accuracy (%, mean ± sample std over 3 seeds) on stratified MNIST subsets. Bold marks the better of the two near-parameter-matched heads (MLP vs. quantum).

			Quantum (re-uploading schedule)		
Train size	Linear	MLP	NONE	LIGHT	STRONG
250	61.5 ± 1.7	36.7 ± 5.4	**54.5 ± 3.7**	48.3 ± 10.7	38.3 ± 7.4
500	79.3 ± 2.9	53.2 ± 6.9	**71.2 ± 1.9**	57.8 ± 2.5	45.6 ± 6.7
1,000	89.6 ± 0.7	76.4 ± 3.4	**83.3 ± 0.3**	68.4 ± 3.5	54.6 ± 14.0

Table 2: Best test-split accuracy (%, mean ± sample std over 3 seeds) on stratified CIFAR-10 subsets. Bold marks the better near-parameter-matched head.

			Quantum (re-uploading schedule)		
Train size	Linear	MLP	NONE	LIGHT	STRONG
1,000	57.5 ± 0.8	**51.2 ± 0.8**	37.1 ± 2.4	34.2 ± 0.7	26.6 ± 1.9
2,000	64.6 ± 0.7	**60.6 ± 0.9**	46.8 ± 5.2	42.7 ± 3.9	28.7 ± 1.9
4,000	71.2 ± 0.5	**67.2 ± 0.4**	58.9 ± 4.6	52.0 ± 7.8	35.2 ± 5.5

CIFAR-10: the ordering inverts. On CIFAR-10 (Table 2) the same quantum head loses to the same near-matched MLP at every training-set size, by 14.1, 13.8, and 8.3 points. Against the linear head the deficit is 20.4 points at 1,000 samples and 12.3 points at 4,000: the quantum head narrows both gaps as data is added but remains far behind, whereas on MNIST it was within 8 points of the linear head everywhere. Training dynamics rule out simple non-convergence: test-split accuracy moved by less than one point over the final ten epochs in all CIFAR-10 configurations.

Generalization-gap analysis separates the failure modes. At 4,000 CIFAR-10 samples the final-epoch train–test gaps are 10.3 (linear), 6.9 (MLP), 5.5 (quantum NONE), and 0.8 points (quantum STRONG). The classical heads overfit; the plain quantum head fits moderately; the strong re-uploading circuit *underfits*: its training accuracy barely exceeds its test-split accuracy at 35%. The VQC is not memorizing and failing to generalize; it lacks the capacity, under this encoding, to fit CIFAR-10's bottleneck features at all. This is consistent with viewing the fixed encoding as a severe architectural prior: a Fourier feature map of bounded frequency content [6] that MNIST embeddings happen to satisfy and CIFAR-10 embeddings do not.

The encoding saturates on CIFAR-10. Direct inspection of trained checkpoints supports the encoding-constraint account. Passing the test split through the backbone of a trained 4,000-sample CIFAR-10 quantum run, 16.7% of bottleneck activations fall in the saturated region of the encoding ($|\tanh z| > 0.9$) and 5.0% are essentially pinned ($|\tanh z| > 0.99$); the corresponding MNIST run yields 4.3% and 0.0%. Saturated dimensions receive vanishing gradient through the encoding and collapse distinct feature values onto nearly identical rotation angles, so the effective resolution of the quantum head degrades exactly on the dataset where more resolution is needed.

Data re-uploading hurts monotonically. Across all six dataset/size conditions, accuracy orders strictly as NONE > LIGHT > STRONG. On CIFAR-10 at 2,000 samples the strong schedule costs 18.1 points relative to no re-uploading. Re-encoding the input before every layer repeatedly rotates the state back toward a data-determined manifold, discarding the entangled representation built by earlier variational layers; with only 4 layers and 8 qubits the circuit apparently cannot compensate. We note this contrasts with the single-qubit and low-dimensional settings where re-uploading provably adds expressivity [5]: expressivity gains at the function-class level do not automatically survive joint training through a CNN backbone.

Seed variance as a trainability signal. Classical heads are nearly deterministic across seeds on CIFAR-10 (sample std ≤ 0.9 throughout), while quantum heads vary by up to 7.8 points. With initialization already controlled (Section 2), the residual variance indicates a rugged optimization landscape with multiple basins, which is mild evidence of the trainability pathologies predicted for VQCs [7], even at a depth and width far below where barren plateaus formally set

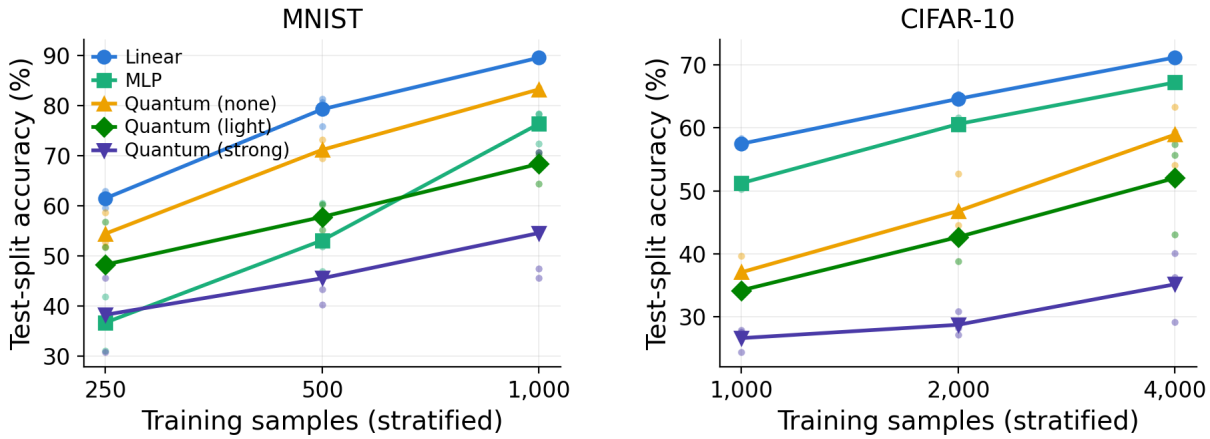


Figure 1: Best test-split accuracy vs. stratified training-set size. Solid lines show the mean over 3 seeds; faint dots show individual seeds. Left: MNIST. Right: CIFAR-10 (legend shared; note the different vertical scales). The quantum head (NONE) sits between the linear and MLP heads on MNIST but below both on CIFAR-10.

in. Optimization difficulty is therefore present, but it is not the dominant effect: even the best seed of the best quantum configuration falls well short of the mean linear-head accuracy on CIFAR-10.

4 Discussion and Limitations

Interpretation. The two datasets bracket the regime where a small VQC head is useful, and the honest summary is conditional: the VQC can outperform a similarly small MLP on MNIST, but it never outperforms a simpler linear probe, and on CIFAR-10 it underperforms both classical heads. When the backbone can compress the task into a nearly linearly separable 8-dimensional embedding (MNIST), the quantum head converts its 154 parameters into substantially more accuracy than a near-matched MLP; the entangling structure acts as a good inductive bias. When the task saturates the bottleneck (CIFAR-10), the fixed rotation encoding becomes the binding constraint, and the saturation measurements above show the mechanism directly: a classical head can reweight and recombine noisy embedding dimensions arbitrarily, while the VQC commits each dimension to a rigid geometric role. The practical implication for hybrid design is that the quantum head's viability is decided upstream, by whether the classical compression stage can deliver an embedding matched to the encoding, not by circuit depth or re-uploading schedule.

Limitations and planned experiments. All quantum results are noiseless statevector simulations; hardware noise would likely further reduce quantum-head accuracy. The evaluation protocol uses the test split for checkpoint selection (Section 2); the measured effect is small (≤ 1.3 points) but a proper train/validation/test split is the correct fix. The backbone is trained jointly with each head, so head comparisons include any differences in the embedding each head induces; a frozen-backbone experiment, training all heads on identical fixed embeddings, is the highest-priority follow-up. Two further experiments would sharpen the encoding-constraint claim: a classical Fourier-feature baseline (embedding $\rightarrow$ sin/cos features $\rightarrow$ linear readout) to test whether the VQC's behavior is explained by its encoding rather than anything quantum-specific, and encoding ablations (learned scale before tanh, clipped angles, normalized embeddings) to test whether relieving saturation closes the CIFAR-10 gap. We fixed the bottleneck at $d = 8$ qubits and depth at $L = 4$; scaling either is natural but simulation cost grows quickly. Finally, three seeds per cell bound, but do not precisely estimate, the seed variance; per-seed results are shown in Figure 1.

5 Conclusion

In a controlled comparison spanning 90 training runs, an 8-qubit VQC classifier head outperforms a near-parameter-matched MLP on MNIST but loses to it on CIFAR-10, and it never outperforms a smaller linear probe on either

dataset. The CIFAR-10 deficit is driven by underfitting linked to saturation of the fixed rotation encoding, rather than by overfitting or simple non-convergence, and data re-uploading reduced accuracy in every configuration tested. These results argue that empirical claims of quantum advantage in hybrid classifiers should be qualified by dataset complexity and accompanied by parameter-matched *and* simple linear baselines. They also locate the frontier for small quantum heads at the quality of the classical embedding they are asked to read.

Reproducibility. Code, sweep configurations, and per-epoch metrics (including per-component gradient norms) for all 90 runs are available at `https://github.com/Arkaeyuh/quantcnn`. Experiments used Python 3.14, PyTorch 2.11, and PennyLane 0.44 (`lightning.qubit`, adjoint differentiation) on an Apple M3 Max; quantum circuits were simulated on CPU with classical components on the MPS backend. Seeds $\{0, 1, 2\}$ control Python, NumPy, and PyTorch RNGs as well as subset selection.

References

[1] M. Benedetti, E. Lloyd, S. Sack, and M. Fiorentini. Parameterized quantum circuits as machine learning models. *Quantum Science and Technology*, 4(4):043001, 2019.

[2] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3:625–644, 2021.

[3] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii. Quantum circuit learning. *Physical Review A*, 98:032309, 2018.

[4] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, and J. M. Gambetta. Supervised learning with quantum-enhanced feature spaces. *Nature*, 567:209–212, 2019.

[5] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J. I. Latorre. Data re-uploading for a universal quantum classifier. *Quantum*, 4:226, 2020.

[6] M. Schuld, R. Sweke, and J. J. Meyer. Effect of data encoding on the expressive power of variational quantum-machine-learning models. *Physical Review A*, 103:032430, 2021.

[7] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9:4812, 2018.

[8] V. Bergholm et al. PennyLane: Automatic differentiation of hybrid quantum-classical computations. *arXiv:1811.04968*, 2018.